SESSION 3

Why AI Ethics Matters

Benefits vs Risks · Responsible AI · Why Location Data Is Different · Global Case Studies · Do & Don't Rules

Day 1 · 7 July 2026 · GeoAI Hackathon Programme

How to use these notes. *These notes are self-contained: they cover everything from the live session plus deeper explanations of the privacy-protection techniques mentioned, expanded case detail, diagrams, a glossary, and self-check questions.*

Learning Objectives

By the end of this session, you should be able to:

- Weigh the real benefits of AI against its real risks, using documented cases on both sides.
- Define Responsible AI and explain why location data raises distinct ethical stakes.
- Explain the three GeoAI-specific ethics risks — re-identification, spatial bias (MAUP), and real-world consequences — and the standard techniques used to manage them.
- Draw practical do/don't rules from three global ethical controversies.
- Complete a Benefits–Risks analysis of a flood early-warning system (Activity 3).

1. Benefits and Risks of AI

AI is neither saviour nor villain — it is powerful, and power cuts both ways. The honest starting point for ethics is to hold the genuine benefits and genuine risks in view at the same time.

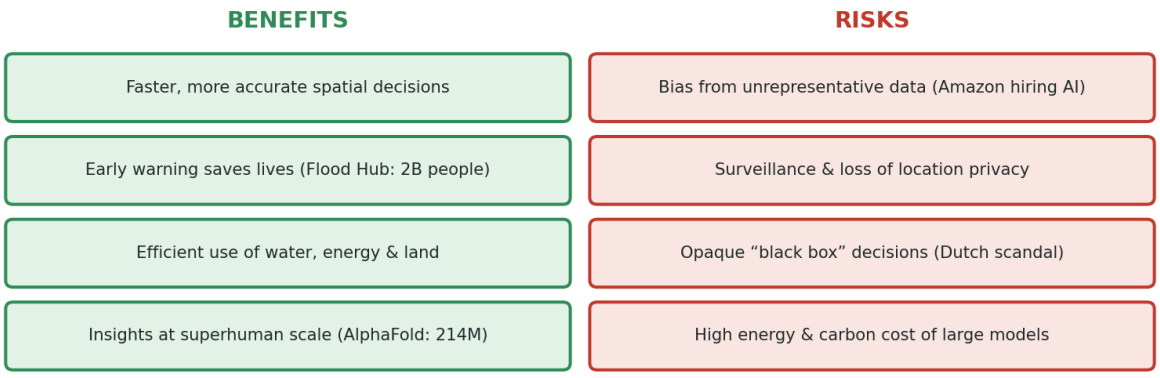


Figure 3.1 — Every benefit on the left has a documented counterpart risk on the right.

Benefits

- **Faster, more accurate decisions from complex spatial data —** AI flags flood-risk zones in seconds from satellite imagery that would take analysts days to review by hand.
- **Early warning for floods, droughts, and disasters —** Google Flood Hub gives 2 billion people up to 7 days' notice before rivers flood (full story below).
- **Efficient use of resources —** AI-guided irrigation cuts water use on a Sabah palm oil estate while protecting yield.
- **Insights at superhuman scale —** AlphaFold mapped 214 million protein structures, work far beyond manual capacity.

Risks

- **Biased predictions from unrepresentative training data —** Amazon's hiring AI penalised résumés containing "women's" after learning from a decade of male-dominated hires.
- **Surveillance and loss of location privacy —** high-resolution satellite monitoring can reveal exactly who is home and when, without consent.
- **Opaque "black box" models —** the Dutch tax authority couldn't clearly explain why its algorithm flagged specific families as fraud risks.
- **High energy and environmental cost —** training one large AI model can emit as much carbon as several cars over their entire lifetimes.

2. When AI Delivers — Two Success Stories

Google Flood Hub (Google Research, since 2018)

Google combined LSTM-based hydrological AI models with satellite imagery and weather data to forecast riverine floods up to 7 days ahead — even in countries with little or no local river-gauge data, closing a warning gap that had existed for decades. This is directly relevant to your hackathon: it proves AI can compensate for sparse ground sensors, exactly the constraint most Malaysian flood projects face.

- 2 billion people now covered by free AI flood forecasts across 150+ countries (2026).
- Up to 7 days' advance warning for riverine floods; up to 24 hours for urban flash floods.

AlphaFold (Google DeepMind, since 2020)

AlphaFold solved a 50-year-old grand challenge in biology — predicting a protein's 3D shape from its amino-acid sequence — then gave the predictions away for free instead of licensing the breakthrough.

- 214 million protein structures predicted and freely published, covering nearly every known protein.
- 3 million+ researchers in 190+ countries used it; the work won the 2024 Nobel Prize in Chemistry.

3. When AI Fails — Two Cautionary Tales

Dutch Childcare Benefits Scandal (Belastingdienst, 2013–2019)

A self-learning fraud-detection algorithm used by the Dutch tax authority treated dual nationality as a risk factor, disproportionately flagging parents with immigrant backgrounds as fraudsters. Flagged families had benefits suspended and full repayments demanded — often tens of thousands of euros — frequently without human review or a workable appeal path.

- ~26,000 families wrongly accused; many forced into debt, some into bankruptcy or worse.
- The 2020 parliamentary inquiry was titled "Ongekend Onrecht" ("Unprecedented Injustice"); in January 2021 the entire Dutch cabinet resigned over the scandal.

Amazon's AI Recruiting Tool (2014–2018, scrapped)

An experimental résumé-screening model was trained on 10 years of past hires in a male-dominated industry. It taught itself to downgrade résumés containing the word "women's" (as in "women's chess club captain") and to favour male-coded language — bias that engineers could not reliably remove, because it was baked into the historical data itself.

- 10 years of gender-skewed historical hiring data baked the bias directly into the model.
- Scrapped in 2017 after Amazon lost confidence it could be made reliably gender-neutral.

Same technology, same era — the difference between the success stories and the failures is not cleverness but safeguards: representative data, human review, transparency, and accountability. Those safeguards are exactly what Sessions 4–6 teach.

4. What Is Responsible AI — and Why GeoAI Needs It

Definition. Responsible AI is the practice of designing, building, and deploying AI so it is fair, transparent, accountable, safe, and privacy-preserving — so the people affected by its decisions can trust it, and challenge it when it gets things wrong.

Link to Session 2: "Ethically Grounded" was named as a core trait of a GeoAI Innovator. Responsible AI is that trait turned into practice — concrete habits and checks that make "weighing risk and fairness" a repeatable process, not just a good intention.

Location data raises the stakes even further. Where you go reveals who you are: where people live, work, worship, seek healthcare, and gather. This "pattern of life" quality makes geospatial AI ethics distinct from general AI ethics. Three risks make this concrete:

Risk 1 · Re-identification

Stripping names and IDs from a location dataset does not make it anonymous — the pattern of where someone goes is often unique enough to identify them on its own, like a fingerprint made of places.

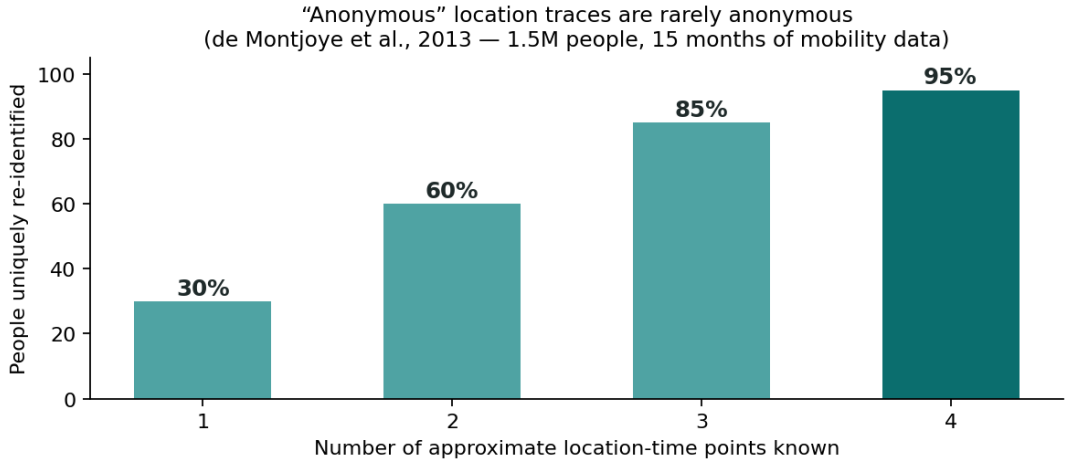


Figure 3.2 — With just 4 approximate location-time points, 95% of individuals in a 1.5-million-person dataset could be uniquely re-identified.

The evidence: MIT researchers studied 15 months of mobility data for 1.5 million people and found that just 4 approximate location-and-time points — with no name attached — were enough to uniquely re-identify 95% of individuals (de Montjoye et al., "Unique in the Crowd", Scientific Reports, 2013).

The standard defences, explained:

- **Aggregation —** publish data only at coarse levels (e.g. per district per day) rather than individual points.
- **k-anonymity —** ensure every record is indistinguishable from at least k−1 others; if any combination of attributes describes fewer than k people, generalise or suppress it.
- **Differential privacy —** add carefully calibrated statistical noise so aggregate patterns remain accurate but no individual's presence in the dataset can be confirmed.
- **Data minimisation —** the simplest defence: don't collect location detail you don't need in the first place.

For a GeoAI innovator: before sharing any location dataset, ask "could 3–4 points here reveal a real person?" — and apply one of the defences above before it leaves your laptop.

Risk 2 · Spatial Bias (MAUP)

The Modifiable Areal Unit Problem (MAUP), first formalised by geographer Stan Openshaw in 1984, says the boundaries and grid size you choose to aggregate data can itself distort the pattern you find — same data, different map, different story. It has two components: the scale effect (results change as your grid cells get bigger or smaller) and the zoning effect (results change when you redraw boundaries at the same scale).

The evidence: gerrymandering is MAUP in action — the same voters, redrawn into different district lines, can flip an election result. In GeoAI terms: a flood-risk model using coarse postcode grids can hide that one street is underwater while its neighbour is dry.

For a GeoAI innovator: test your model's output at more than one spatial resolution before trusting it, and document why you chose your specific boundaries — judges can and do ask.

Risk 3 · Real-World Consequences

Unlike a mis-recommended video, a biased geospatial model can misdirect emergency resources, wrongly deny services, or fail to warn exactly the people who need it most. Geospatial errors have street addresses.

The evidence: in the Dutch childcare benefits case, the algorithm's flags weren't abstract — they triggered real debt, real repossessions, and real family breakdowns for roughly 26,000 households.

For a GeoAI innovator: before demo day, run a "who could this fail?" check — exactly what the Fairness and Human Oversight items on your Hackathon Ethics Checklist (Session 6) are there for.

5. Global Ethical Case Studies — What To Do, and Not Do

Three real controversies, each converted into rules your team can actually follow this weekend.

Case A · Predictive Policing Maps (US)

Location-based crime-prediction tools trained on historical arrest data have been shown to reinforce over-policing in already-disadvantaged neighbourhoods. The core flaw: arrest data records where police looked, not necessarily where crime occurred — so the model learns and amplifies past patrol patterns, creating a feedback loop.

- **DO:** audit training data for historical bias before deployment; pair any risk score with human review and a clear appeal path before action is taken on it.
- **DON'T:** treat historical arrest data as neutral "ground truth"; deploy a location risk score without testing its error rate across different neighbourhoods.

Case B · Satellite Surveillance & Land Rights

High-resolution earth observation has been used to monitor land use in ways that, without consent frameworks, can threaten indigenous and rural land claims — for example, imagery interpreted by outsiders being used as evidence in disputes about land the community has occupied for generations.

- **DO:** set up clear consent and data-use agreements with communities before monitoring their land; involve local or indigenous stakeholders in interpreting what the imagery actually shows.
- **DON'T:** assume public satellite access means unlimited rights to monitor or act on what you see; share land-use findings with governments or companies without the community's knowledge.

Case C · Flood Early-Warning Systems

Life-saving alerts must balance broad public benefit with the privacy of the granular household data used to generate them. This is the closest case to most hackathon projects — and the scenario of Activity 3 below.

- **DO:** aggregate or mask household data to the coarsest resolution that still gets the alert right; be transparent with communities about what's collected and exactly who can access it.

- **DON'T:** collect more granular location data than the warning actually needs, "just in case"; let a system's urgency become an excuse to skip privacy safeguards entirely.

6. Hands-On Activity 3 — Benefits vs Risks Analysis (10 minutes)

Scenario: a flood early-warning system vs privacy concerns. In groups — or on paper if self-studying — identify the opportunities and risks of the application.

1. List the public-safety benefits of the system.
2. List the privacy and data risks it introduces.
3. Rate the overall risk level: Low / Medium / High — and be ready to justify it.

Deliverables: an AI Benefits–Risks Table and a Risk Awareness Matrix.

Benefits	Risks	Risk level
e.g. Earlier evacuation, fewer casualties	e.g. Household data could reveal who is away from home	Medium
Your answer:		

The key facilitator prompt, worth internalising: name who bears the risk versus who gets the benefit — they are not always the same people. A system whose benefits go to one group while its risks land on another has a fairness problem even if the totals look positive.

Glossary

Term	Meaning
Responsible AI	The practice of building AI that is fair, transparent, accountable, safe, and privacy-preserving.
Black box model	A model whose internal reasoning cannot be readily inspected or explained.
Re-identification	Recovering individuals' identities from supposedly anonymised data.
k-anonymity	A privacy standard: every record must be indistinguishable from at least k−1 others.
Differential privacy	Adding calibrated statistical noise so aggregates stay accurate but individuals cannot be confirmed present.
Data minimisation	Collecting only the data a task genuinely requires.
MAUP	Modifiable Areal Unit Problem — analysis results change with the boundaries/grid chosen (scale effect + zoning effect).
Pattern of life	The revealing routine visible in someone's movement history (home, work, worship, healthcare).
False positive / negative	A false alarm (predicting an event that doesn't happen) / a miss (failing to predict one that does).
Feedback loop	When a model's outputs shape the future data it learns from, amplifying its own bias.

Self-Check Questions

1. Why did the Amazon recruiting model become biased even though nobody programmed it to discriminate?

2. Your teammate says: “We removed all names and phone numbers, so the GPS dataset is anonymous.” Using the MIT study, explain in two sentences why this is wrong, and name two techniques that would actually help.
3. Give a concrete example of MAUP’s zoning effect in a flood-risk map.
4. In the predictive-policing case, what is the feedback loop, and which single DO-rule breaks it?
5. For a flood early-warning system, who typically bears the privacy risk, and who receives most of the safety benefit? Are they the same people?

Key Takeaways

- Every AI benefit carries a corresponding risk; the difference between Flood Hub and the Dutch scandal is safeguards, not technology.
- Responsible AI turns good intentions into repeatable checks: fairness, transparency, accountability, safety, privacy.
- Location data is special: 4 anonymous location points can re-identify 95% of people; boundary choices (MAUP) can distort results; and errors have physical, life-and-property consequences.
- The standard privacy defences — aggregation, k-anonymity, differential privacy, data minimisation — are practical enough to apply during a hackathon weekend.
- Always ask who bears the risk and who receives the benefit — they are rarely the same people.

Sources & Further Reading

- Google Research — How Google Uses AI to Improve Global Flood Forecasting (2026); Google DeepMind — AlphaFold: Five Years of Impact (2026).
- Dutch Parliamentary Inquiry “Ongekend Onrecht” (2020); Reuters — Amazon Scraps Secret AI Recruiting Tool (2018).
- de Montjoye et al., “Unique in the Crowd: The Privacy Bounds of Human Mobility”, Scientific Reports (2013).
- Openshaw, “The Modifiable Areal Unit Problem” (1984); Harvard CGA — The Critical Challenges of GeoAI.
- MDPI — An Operational Ethical Framework for GeoAI (2026).